

SCIENCE- Class (6-8)

Curricular Goals and Competencies

Curricular Goals	Competencies
CG-1 Explores the world of matter and its constituents, properties, and behaviour	1.1 Classifies matter based on observable physical (solid, liquid, gas, shape, volume, density, transparent, opaque, translucent, magnetic, non-magnetic, conducting, non-conducting) and chemical (pure, impure; acid, base; metal, non-metal; element, compound) characteristics
	1.2 Describes changes in matter (physical and chemical) and uses particulate nature to represent the properties of matter and the changes
	1.3 Explains the importance of measurement and measures physical properties of matter (such as volume, weight, temperature, density) in indigenous, non-standard and standard units using simple instruments
	1.4 Observes and explains the phenomena caused due to differences in pressure, temperature, and density (e.g., breathing, sinking-floating, water pumps in homes, cooling of things, formation of winds)
CG-2 Explores the physical world in scientific and mathematical terms	2.1 Describes one-dimensional motion (uniform, nonuniform, horizontal, vertical) using physical measurements (position, speed, and changes in speed) through mathematical and diagrammatic representations
	2.2 Describes how electricity works through manipulating different elements in simple circuits and demonstrates the heating and magnetic effects of electricity
	2.3 Describes the properties of a magnet (natural and artificial; Earth as a magnet)
	2.4 Demonstrates rectilinear propagation of light from different sources (natural, artificial, reflecting surfaces), verifies the laws of reflection through manipulation of light sources and objects and the use of apparatus and artefacts (such as plane and curved mirrors, pinhole camera, kaleidoscope, periscope)
	2.5 Observes and identifies celestial objects (stars, planets, natural and artificial satellites, constellations, comets) in the night sky using a simple telescope and images/ photographs, and explains their role in navigation, calendars, and other phenomena (phases of the moon, eclipse, life on earth)
CG-3 Explores the living world in scientific terms	3.1 Describes the diversity of living things observed in the natural surroundings (insects, earthworms, snails, birds, mammals, reptiles, spiders, diverse plants, and fungi), including at a smaller scale (microscopic organisms)
	3.2 Distinguishes the characteristics of living organisms (need for nutrition, growth and development, need for respiration, response to stimuli, reproduction, excretion, cellular organisation) from non-living things
	3.3 Analyses patterns of relationships between living organisms and their environments in terms of dependence on and response to each other

	3.4 Explains the conditions suitable for sustaining life on Earth and other planets (atmosphere; suitable temperature-pressure, light; properties of water)
CG-4 Understands the components of health, hygiene, and wellbeing	4.1 Undertakes a nutrition-based analysis of food components with special reference to Indian culinary practices and modern understanding of nutrition, and explains the effect of nutrition on health
	4.2 Examines different dimensions of diversity of food — sources, nutrients, climatic conditions, diets
	4.3 Describes biological changes (growth, hormonal) during adolescence, and measures to ensure overall well-being
	4.4 Recognises and discusses substance abuse, viewing school as a safe space to raise these concerns
CG-5 Understands the interface of Science, Technology, and Society	5.1 Illustrates how Science and Technology can help to improve the quality of human life (health care, communication, transportation, food security, mitigation of climate change, judicious consumption of resources, applications of artificial satellites) as well as some of the harmful uses of science in history
	5.2 Shares views on news and articles related to the impact that Science/Technology and society have on each other
CG-6 Explores the nature and processes of Science through engaging with the evolution of scientific knowledge and conducting scientific inquiry	6.1 Illustrates how scientific knowledge and ideas have changed over time (description of motion of objects and planets, spontaneous generation of life, number of planets) and identifies the scientific values that are inherent and common across the evolution of scientific knowledge (scientific temper, Science as a collective endeavour, conserving biodiversity and ecosystems)
	6.2 Formulates questions using scientific terminology (to identify possible causes for an event, patterns, or behaviour of objects) and collects data as evidence (through observation of the natural environment, design of simple experiments, or use of simple scientific instruments)
CG-7 Communicates questions, observations, and conclusions related to Science	7.1 Uses scientific vocabulary to communicate Science accurately in oral and written form, and through visual representation
	7.2 Designs and builds simple models to demonstrate scientific concepts
	7.3 Represents real world events and relationships through diagrams and simple mathematical representations
CG-8 Understands and appreciates the contribution of India through history and the present times to the overall field of Science, including the disciplines that constitute it	8.1 Knows and explains the significant contributions of India to all matters (concepts, explanations, methods) that are studied within the curriculum in an integrated manner

CG-9 Develops awareness of the most current discoveries, ideas, and frontiers in all areas of scientific knowledge in order to appreciate that Science is ever evolving and that there are still many unanswered questions	9.1 States concepts that represent the most current understanding of the matter being studied — ranging from mere familiarity to conceptual understanding of the matter as appropriate to the developmental stage of the students
	9.2 States questions related to matters in the curriculum for which current scientific understanding is well recognised to be inadequate